

MICHELLE POLINKA
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SUMMARY

Full-stack software engineer with an M.S. in Computer Science and Engineering, currently building AI-powered systems. Experienced in LLM integration, RAG, production APIs, and real-time data systems. Proficient in Python, React, TypeScript, and SQL.

EDUCATION

M.S. Computer Science and Engineering: University of California, Santa Cruz

- September 2025 – June 2026

B.S. Computer Engineering: University of California, Santa Cruz

- September 2022 – June 2025

EXPERIENCE

IpserLab – Software Engineer: July 2026 – Present

- Engineered secure authentication with FastAPI, Argon2, JWTs, and Pydantic
- Evolved SQLAlchemy schemas through backward-compatible Alembic migrations
- Resolved a React and TypeScript defect involving nullable expiration dates
- Diagnosed OAuth configuration issues across frontend and backend environments
- Validated changes through pytest regression suites and GitHub code reviews

UCSC Faculty-Led Team – AI Software Engineer: June 2026 – July 2026

- Architected a citation-grounded RAG pipeline for reliable academic advising responses
- Developed context-aware intent orchestration for precise multi-turn course and policy queries
- Implemented provenance-preserving retrieval across prerequisite, catalog, schedule, and availability data
- Integrated a Neo4j degree-audit engine using deterministic, rule-based academic evaluation
- Established reproducible Docker and CI workflows spanning application, graph, and data-integrity testing

Parlo – Software Engineer: September 2025 – February 2026

- Engineered a real-time Firebase/Firestore synchronization layer supporting 150+ concurrent active users
- Redefined Firestore data models and batch write workflows to eliminate race conditions across clients
- Optimized database query patterns and indexing strategies, contributing to a 17% increase in daily active users
- Orchestrated internal tooling and monitoring solutions to identify system bottlenecks and accelerate debugging
- Designed scalable backend architectures supporting rapid feature development, reliability, and performance

BiometerPlus – Software Engineer Intern: June 2024 – September 2024

- Developed RESTful API endpoints with MySQL integrations supporting biometric data collection and retrieval
- Built MySQL-backed authentication and account management systems validated through Postman test suites
- Implemented full account lifecycle management with soft deletion, reactivation, and JSON schema validation
- Constructed a secure password recovery workflow using time-limited tokens and server-side validation checks
- Enhanced backend reliability through input validation, authentication controls, and structured error handling

PROJECTS

JazzyFS: Adaptive Experimental Filesystem (Master's Thesis)

- Implemented a FUSE-based filesystem with adaptive file prefetching and runtime access analysis
- Created instrumentation and benchmarking tools to evaluate filesystem performance across multiple workload types
- Integrated a sonification system converting filesystem activity into real-time audio feedback
- Created automated experiment pipelines for repeatable testing, performance analysis, and result generation

Fault-Tolerant Distributed Sensor Network

- Developed a fault-tolerant token-ring protocol across three Raspberry Pi devices
- Implemented reliable data transfer and state synchronization between network nodes
- Built automated health checks to detect inactive or unresponsive devices
- Created real-time failure notifications to alert operators when outages occurred

TECHNICAL SKILLS

- **Languages:** Python, C++, TypeScript, SQL
- **Frameworks and Libraries:** React, FastAPI, SQLAlchemy, Alembic, pytest, Pydantic, FUSE
- **AI/ML:** LLM integration and tool-calling, retrieval-augmented generation (RAG), vector search
- **Databases:** PostgreSQL, MySQL, Firebase/Firestore, Neo4j
- **Tools and Infrastructure:** Linux, Docker, Docker Compose, Git, GitHub Actions, Vitest, Bash, Postman

ADDITIONAL

- Advanced Certificate of Merit in Classical Piano
- Composed and copyrighted original music